

Varunprasaath R. Selvaraju, Ph.D.

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PROFESSIONAL SUMMARY

PhD chemist/materials scientist with 6+ years in process development, materials characterization, and failure/root-cause analysis across polymers, thin films, and functional molecular materials. Applies DOE/SPC, SOP/control-plan thinking, and multi-technique analytical readouts to stabilize high-variation R&D workflows and define reproducible process windows.

CORE COMPETENCIES

Process Engineering

DOE/SPC

Root Cause Analysis

Materials Characterization

Workflow Control

Lab Operations

Data Integrity

WORK EXPERIENCE

University of Colorado Boulder

July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided bio-based polymer process development, defining process inputs/outputs, material-quality specifications, and reproducible operating windows for scale-up.

University of Colorado Boulder

July 2021 - July 2025

Graduate Researcher

- Assessed failure modes and structure-property-performance relationships across 15+ functional material variants using spectroscopy, electrochemistry, thermal analysis, chromatography, and MS.

Georgia Institute of Technology

Aug 2019 - July 2021

Graduate Researcher

- Drove thin-film process development for AFOSR-funded coating systems, translating surface-preparation and CVD variables into controlled morphology and optical performance.

ETH Zurich

May 2017 - July 2017

Research Assistant

- Developed reproducible CVD polymer thin-film workflows with DOE and in-situ QCM-D monitoring, connecting equipment conditions to growth kinetics and film quality.

University of Colorado Boulder

2021 - 2025

Laboratory Operations Manager

- Managed characterization/process-development lab operations for a 15-member group; implemented calibration, SOP, and safety systems that improved instrument uptime by 30%.

Georgia Tech

2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Bio-Based Polymer Scale-Up Process Current

Stabilized process variables and material-quality readouts for reproducible polymer scale-up using DOE/SPC and technical documentation.
DOE, SPC, process windows, characterization

Electrochromic Material Reliability Study Chemistry of Materials 2025

Mapped failure modes, degradation behavior, and performance across functional materials with orthogonal characterization.
FT-IR, UV-Vis-NIR, CV, TGA/DSC, NMR, GPC, MS

AFOSR Defense Thin-Film Coatings DoD Funded

Built controlled thin-film workflows by connecting CVD process variables to morphology, interfaces, and optical performance.
CVD, QCM-D, surface prep, thin films

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder

2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials
Advisor: Prof. Seth Marder
(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc)

2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Process Engineering: DOE, SPC, process windows, root cause analysis, control plans, SOPs, technical reports, reproducibility studies
Characterization: FT-IR, UV-Vis-NIR, ellipsometry, QCM-D, TGA, DSC, NMR, GPC/SEC, LC-MS, GC-MS, cyclic voltammetry
Materials: Polymers, organic/organometallic materials, thin films, coatings, surface functionalization, structure-property-processing relationships
Data & Statistics: JMP (basic), Python (basic), Excel, structured experimental records, trend analysis, data integrity
Operations: Lab operations, equipment tracking, calibration systems, safety/SOP systems, cross-functional resource planning

JOURNAL PUBLICATIONS

- Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." *Chemistry of Materials*, 2025, 37, 5558-5568.
- Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." *RSC Advances*, 2018, 8, 20048-20055.
- Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.
Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.
Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.
Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)